

Preparing Children with Level 1 ASD for Social Interactions through Storytelling with Amy: An Exploratory Study

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ABSTRACT

Technology-based interventions can help children with autism spectrum disorder (ASD) prepare for social interactions by addressing their limited social communication and emotion regulation skills. In particular, chatbots' natural language abilities can provide a conversation-based interaction that is safe and individualized. This motivates us to design Amy as a virtual talking companion that delivers social stories to children with Level 1 ASD. Through a week-long experiment, we captured the perception and feedback of four children and their parents on Amy's story-based conversations using performance, humanity and affect as the evaluation metrics. Children found the story themes to be relevant and the accompanying visuals to be helpful in their understanding of the stories and their engagement with Amy. Parents and child educators praised Amy for being considerate of the children's feelings. Large scale experiments can further shed insights on the utility of a chatbot that anchors its delivery on social-emotional learning theories.

CCS CONCEPTS

• **Human-centered computing** → **Natural language interfaces**; *Interaction paradigms*; *Interaction design*.

KEYWORDS

conversational agent, social-emotional learning, storytelling, autism spectrum disorder

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1 INTRODUCTION

Autism spectrum disorder (ASD) is characterized by deficits in social communication, social interaction and emotion regulation [23, 24].

Among the three levels of severity identified by the American Psychiatric Association [2], Level 1 needs the least support during social settings to address social impairments caused by difficulty in initiating social interaction and responding to social overtures of people. Individuals in this level may also encounter challenges in regulating their emotional expression and therefore may overreact negatively to relatively minor incidents [15].

Prior studies have investigated the use of conversational agents (CA) to stimulate social interaction and emotional engagement of children with ASD. For instance, the robot system Leo combined conversational skills with robotic movement to facilitate the child-robot interaction [23]. A chatbot model identical to Amazon's Alexa and Google Assistant learns through the interaction to render a more personalized approach that adjusts to the social capacity of the child [25]. The virtual peer Elli / Ellie embedded in an interactive storytelling system utilizes social stories as it guides children with ASD to safely navigate social situations in a virtual environment [6]. FaceMe uses a virtual agent in its augmented reality system to teach basic emotions while improving the emotional and communication skills of children with ASD [17]. Live Interactive Social Skills Assistance (LISSA) provides an online environment where teens with ASD can hold private conversation practice to develop their social skills [1].

Storytelling can be a powerful aid in helping children make sense of social situations with the added benefit of enriching their vocabulary [9]. Carol Gray [13] introduced social stories to help individuals with ASD understand and face situations that may trigger stress and anxiety. Social stories describe a particular situation and guide children in acquiring a proper rule of conduct that they can apply in their everyday social situations [3]. When used appropriately, social stories can prevent, manage, and eliminate aggressive behavioral reactions [5].

Most studies on the use of CAs by individuals with ASD are limited to addressing either social skill development or emotion regulation. In our study, we investigate how a CA designed as a virtual talking companion leverages social stories as its communication strategy and combines this with emotion regulation facility to assist children with Level 1 ASD. Amy, as we call it, delivers social stories using various modalities – text, voice and visual images. Throughout the interaction, Amy remains keen on monitoring the child's emotion and facilitates the delivery of emotion regulation intervention as needed. End-of-story assessment allows Amy to evaluate the child's recall and comprehension of the shared story. The design of Amy was based on a series of interviews with parents and child psychologist leading to a combined features comprising

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of relevant social story themes, open-domain conversation flow, guided emotion regulation exercise, and lively visual interface to complement the text-based interaction.

Through a week-long experiment, we captured the perception and feedback of four (4) children and their parents on Amy's social-story-themed conversations using performance, humanity and affect as the evaluation metrics. Results showed that children found the story themes to be relevant and the accompanying visuals to be helpful in their understanding of the stories and their engagement with Amy. Parents and child educators, on the other hand, praised Amy for being considerate of the child's feelings.

2 DESIGNING AMY

Amy is a virtual talking companion that applies social-emotional learning theories [7] to help children who are 5 to 11 years old and with level 1 ASD recognize and manage their emotions, prepare for social situations, and establish positive relationships with others. As a storyteller, Amy uses a narrative-based conversation flow depicted in Figure 1 to explain social situations and describe appropriate responses that model proper behaviors. Speech is generated using Google's Text-to-Speech service. Oral communication is one-way which means the child responds to Amy by typing free-form text input or selecting one of the given quick replies (Figure 2) in the chat interface. Amy communicates verbally and in textual form using the English language which is one of the official languages used by the government and educational institutions throughout the country.

Parents, caretakers and a child psychologist were part of the consultative process in designing Amy. Specifically, individual interviews with five (5) parents of children with Level 1 ASD and one (1) child psychologist were conducted to identify challenges in social interactions faced by the children, exposure of parents and children to social stories and conversational agents, and access to gadgets and learning applications [10]. Thematic analysis of the interview data led to the formulation of 12 story themes, the construction of a narrative-based conversation flow (Figure 1) that is both informative and interactive to deliver lessons on social situations, and the selection of an appropriate emotion regulation exercise as an intervention when a negative emotion is detected during the bot-child interaction.

The 12 stories cover themes related to everyday social situations that children may encounter and include *greetings, respecting personal space, taking turns, helping others, asking for help, listening to others, and dealing with feelings of comfort and discomfort*. Additional themes related to health, specifically *staying healthy, limiting use of gadgets, grooming, and social distancing in the context of COVID-19*, as advised by the psychologist, complete the line-up. The stories were constructed following Carol Gray's guidelines [14] and contextualized to the background of the target demographics that were derived from the interviews with stakeholders. To enhance reading comprehension, visual cues in the form of simple and realistic pictures representing story characters, objects, setting and events are utilized to complement the story text [19]; an example is illustrated in Figure 3.

As a tutor, Amy poses comprehension questions to assess children's recall and understanding of the story events and proper

behavior during the *Assessing* phase of the conversation. In the *Evaluating* phase, Amy generates one of six (6) types of responses following the dialogue moves of AutoTutor [11] and Eren [21]. These include *feedback* on the correctness of the answer; *hint* where the child can choose from three (3) candidate answers; *correcting* to give the right answer to a given question; *pumping* the child to express his/her emotion ("Would you like to tell me how you feel?"); *reflecting* on the events experienced by the child ("I know it's hard, but do you think what happened was right?", "What do you think should have been done instead?"); and *summarizing* the main events of the story to reinforce learning.

Amy incorporates the emotion detection of Eren [21] to monitor the child's emotions during the Labelling phase. The following 14 emotions can be identified: joy, gratitude, satisfaction, love, relief, surprise, disappointment, distress, shock, sorry for, fear, resentment, anger, and hate. If a negative emotion is detected, Amy executes its emotion regulation (ER) facility that involves a Sharing phase, a Reflecting phase and an ER exercise. Since Eren's detection facility does not always accurately identify the emotion of the child, Amy adapts the emoticons from the Anger Scale by Buron and Curtis [4], as shown in Figure 4, to help children with ASD to be more aware of their emotions.

3 METHOD

Participants. Four (4) children with Level 1 ASD participated in our week-long experiments. Two of the children were 11 years old while the other two were age 5 and 7 years old. Parents and children duly accomplished informed consent forms and informed assent forms, respectively.

Procedure. The participants were asked to use Amy at least twice a day for five (5) days under the supervision of their parents. The conversations between the children and Amy were recorded and anonymized for later analysis by the child psychologist. Parents were asked to answer two survey forms on behalf of their children: the *pre-intervention survey* and the *post-intervention survey*.

The *pre-intervention survey* collects information regarding the children that includes basic demographics (*name, age, gender*), prior experience with chatbots, exposure to social stories, and exposure to storytelling using a 5-point Likert scale representing *never, rarely, sometimes, often* and *always*. No screening was conducted to determine the participants' literacy skills in reading and writing and their comprehension skills in writing-listening.

The *post-intervention survey* collects parents' perception on their children's enjoyment and satisfaction during and after their interaction with Amy using three criteria – *performance, humanity* and *affect* adapted from [20]. *Performance* measures the appropriateness and correctness of Amy's narration and comprehension questions, assessment of children's answers, and emotion detection and regulation. *Humanity* examines Amy's ability to converse naturally with the child, while *affect* assesses the child's enjoyment and satisfaction in interacting with the virtual talking companion.

Analysis. The experiments yielded a total of 32 conversation logs. These logs were analyzed to understand the children's behavior during their interaction with Amy and to assess the overall performance of Amy.

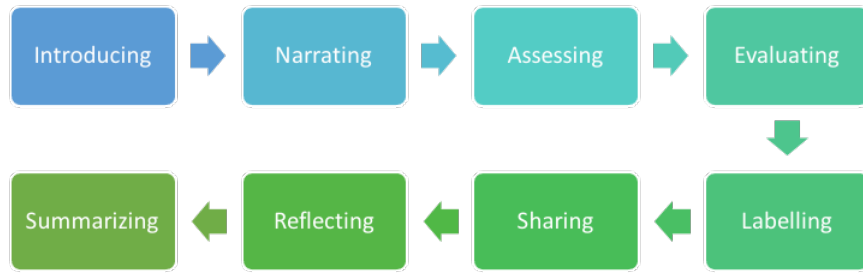


Figure 1: Phases in the conversation flow of Amy.

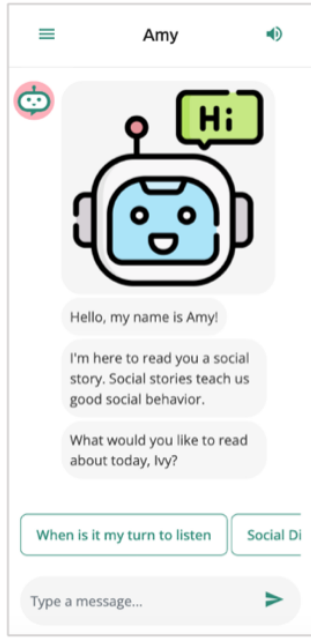


Figure 2: Options for quick replies

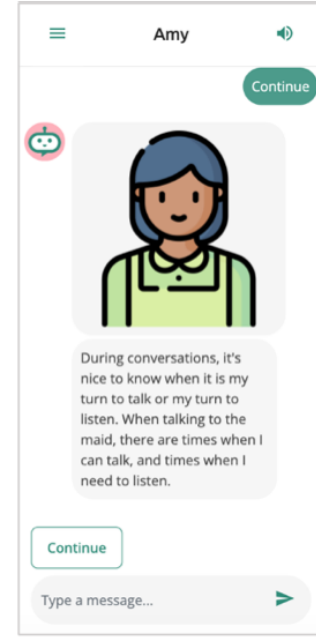


Figure 3: Story text with picture

4 RESULTS AND DISCUSSION

We clustered our results into two - children's performance and Amy's performance - to determine how children responded to Amy's storytelling and to evaluate Amy's performance of its tasks.

4.1 Children's Performance

Results from the pre-intervention survey revealed that only one (1) of the four (4) participants has had a prior experience with a chatbot. Three (3) participants have read or listened to a social story from their therapists, parents, television and social media sites.

Each session may include the delivery of one or more stories, assessments, and emotion regulation exercises. Figure 5 shows the participation rate of children based on the number of sessions they had with Amy, and the number of completed stories and assessments. The participation rate is indicative of how the design of Amy is able to motivate the children to finish listening to the stories and to answer the comprehension questions that follow. *Child A*, who has the least number of sessions, achieved 100% completion

rate in listening to the stories and answering the comprehension assessments. However, *Child B* who has the next highest number of completed stories also has the lowest number of completed assessments. As this participant is still learning how to read, he/she encountered difficulty in understanding the comprehension questions, thus he/she ended most of the assessments when Amy posed a question that he/she could not answer.

We performed quantitative analysis on the conversation logs by measuring the different types of answers children gave to Amy's assessment questions. This is illustrated in Figure 6 where the y-axis represents the number of occurrences while the x-axis represents the frequency of each type of answers vis-a-vis the assessment questions. Amy generates a wait-time feedback ("Do you need more time to think of an answer?") when the child has been silent for 30 seconds after a question has been raised. The total number of correct and incorrect answers garnered by the children are also indicated. We found from the logs that there are instances when Amy did not accept a logically correct answer because the keyword was misspelled or is not found in Amy's knowledge base, or the

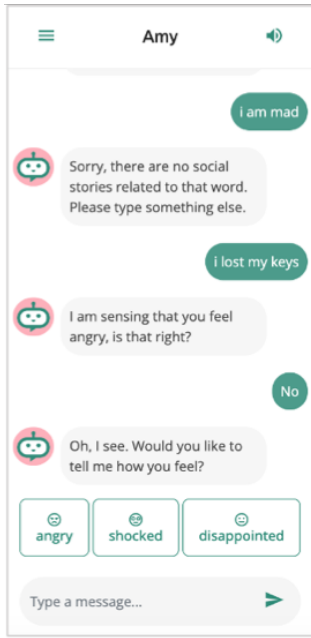


Figure 4: Anger scale

given answer has a different meaning. Table 1 illustrates instances of these cases.

4.2 Amy's Performance

Table 2 shows the average evaluation scores that Amy received from the post-intervention survey. For the *performance* criterion, Amy has an average rating of 4.05. Children generally knew how to start a conversation with and to respond to Amy's questions accordingly. The chatbot can guide the children in performing activities and provide relevant information regarding the stories using a language that is easy to understand. Not only do the social stories contain themes that are relevant to the present-day context but the consistent display of pictures played a big role in helping these children who are visual learners comprehend the stories. Mixed results, however, were received for assessments; some children found the

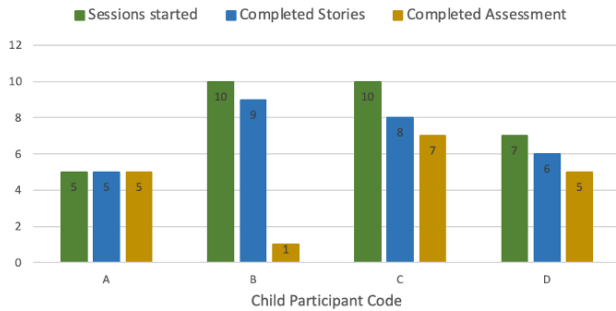


Figure 5: Total number of sessions vis-à-vis children's completion rates in listening to the narrated stories and answering the assessment questions.

Table 1: Examples of logically correct answers that Amy tagged as incorrect due to (a) a misspelled keyword, (b) the keyword is not found in the knowledge base, or (c) the given answer is not the expected answer.

Conversation (a)		
1	Amy	What will we do when it is 9:00 AM?
2	Child C	Exercise
3	Amy	Good try. But the answer could be we will exercise.
Conversation (b)		
1	Amy	What will others feel when we keep on interrupting them?
2	Child A	Disappointed
3	Amy	Good try. But the answer could be disappointment. They would feel this way because we are disrespectful.
Conversation (c)		
1	Amy	What will we do when we are using the book and our teacher asked us if we could borrow it?
2	Child C	Let him borrow
3	Amy	Nice try. But the answer could be asking our teacher to play with the book together.

comprehension questions to be very easy while others encountered challenges answering the questions, such as *Child B*. While the breathing exercise was found effective in helping children regulate their emotion, the same is not true for the sharing and reflection session, thus, the average score of 3.92 for the emotion detection and regulation component.

In the humanity aspect, there were conflicting opinions on how natural and comfortable it is to converse with Amy. This is attributed to the sound of the chatbot's voice; one child disagreed with the statement that "*Amy sounds human*" while another stated that "*the sound [of Amy's voice] is [like] a natural speaking voice*". A parent also recommended that Amy should "*have feelings*" as it speaks and to share short stories in Filipino.

Affect has the lowest average rating among the three criteria. Most of the questions were answered neutrally by half of the participants. The children enjoyed listening to the stories the most but have mixed feelings on sharing their experiences and emotions with Amy. A parent recalled that their child "*started off not wanting to do it [listen to Amy] but eventually was able to sit through the social stories*". Another parent praised Amy's storytelling as it helped their child "*remember the stories*". Moreover, Amy's feedback was able

Table 2: Average evaluation scores of Amy derived from the post-intervention survey that were answered by the parents of the participants using a 5-point Likert scale. The scores for the components of the Performance criterion are also indicated.

Performance	4.05	Overall interaction	4.18
		Narration of social stories	4.19
		Conduct of assessments	3.92
		Emotion detection and regulation	3.92
Humanity	3.69		
Affect	3.67		

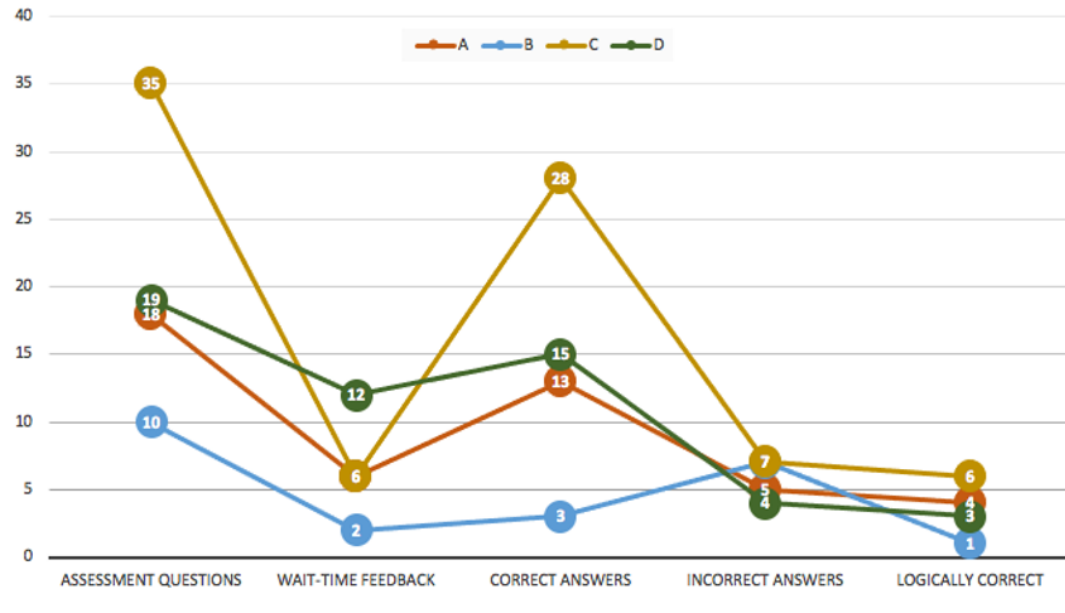


Figure 6: Children’s wait-time feedback vis-à-vis correctness of answers during the assessments.

Table 3: Excerpts of Amy’s responses containing vocabulary terms and emotions that may be difficult for young children to understand.

- 1 I’m here to read you a social story. Social stories teach us **good behavior**.
- 2 If I do things that make other people feel uncomfortable, they might dislike me and **refuse** to play with me.
- 3 Using gadgets can have both **positive and negative** effects on me.
- 4 I will show expected behaviors to make others around me feel **comfortable and joyful**.

to make the children feel validated regardless of the correctness of their answer.

Two experts – a psychologist and a behavioral therapist – analyzed the chatlogs and offered their views on Amy. They found instances of story text containing vocabulary concepts and emotions that are too advanced for and may not be familiar to young children. Examples of these terms are shown in boldface italics font in Table 3. According to the experts, this may extend to comprehension questions that are challenging to answer because children with ASD struggle with abstract concepts such as *comfortable*, *joyful* and *pleasant*. The children may end up just repeating the words of Amy without understanding its meaning, as one expert observed from the chatlogs: “*Note that the child possibly just repeated what was stated. I am doubtful that he/she knows what joyful truly means.*” Instead, Amy should exemplify concrete answers which the children can model, such as “*Give others a smile when we see them.*”

4.3 Design Implications

Children with ASD are vulnerable in social situations due to their limited communication and emotion regulation skills. While a conversational agent may provide a safe space for social-emotional learning to take place, it is crucial to consider their specific needs when designing support-based systems to prevent feelings of distress, distraction and demotivation. The following considerations were made in the development of Amy: selecting the story themes, designing the chat interface, formulating the dialogue templates, selecting an appropriate ER exercise, and assigning Amy’s roles.

Selecting the story themes. Story themes should center on social situations that children with ASD may encounter in their everyday routines. The goal is not only to help them learn the proper behavior and responses in such situations, but also to make learning more meaningful and engaging. Aside from common situations where practices on respecting personal space, sharing, helping and listening may be applied, depicting the current context brought about by the COVID-19 pandemic was also necessary to teach children how to take precautionary measures during emergency situations in order to safeguard themselves and others. This was explicitly highlighted by the expert.

Results from the post-intervention survey revealed that the children enjoyed listening to stories with socially relevant themes. Similarly, the experts agreed that the story themes were appropriate for the target audience and that the lessons contained in the stories will be helpful in developing the social skills of the children. Encouraged by the positive responses of their children when listening to Amy’s stories, parents suggested increasing the number and variances of the stories and to deliver stories in other languages, such as the national language Filipino.

Designing the chat interface. A free-form chat interface gives children the opportunity to respond to Amy’s questions using their

own words. However, there are instances when children may be stuck and would require some assistance in responding to Amy. The availability of quick replies that provide children with options to choose from is one mechanism for the chatbot to aid those who may encounter challenges in formulating an answer.

It is also important to note that children with ASD are highly visual. Inputs from consultations with parents led to the inclusion of pictures that depict story elements to help children comprehend the narratives. However, to prevent sensory overload, muted earth tones were used as the overall color scheme [12, 22]. This decision did not merit any negative feedback from the children and their parents. Furthermore, the children found the pictures that accompanied the story text to be helpful. When asked about what they liked the most in their interaction with Amy, three children responded with “the pictures”. These support the claim of More [19] where pictures and illustrations that represent story elements can aid in increasing children’s understanding of the story.

Formulating the dialogue templates. A sensitive word, incorrectly phrased statement or a vague instruction may trigger emotional distress that can disrupt the learning activity of the child. Proper word choices and the use of concise language are important considerations when designing interactive systems for children with limited communication abilities. Concise statements can convey messages in short statements that are easy to read and understand.

Evaluation from parents and children revealed that Amy’s responses are easy for the children to understand and the corrective feedback to incorrect answers are not causing discouragement. However, the experts recommended that “[Amy’s responses] can further be improved in terms of vocabulary and length of sentences.” They expressed concern over how “the vocabulary seemed a bit too complicated, especially for the younger children in the given age bracket, to understand” and that “some sentences are too long”.

Selecting an emotion regulation exercise. Every action, from listening to the stories to answering the comprehension questions, expressing one’s emotions and reflecting on proper behavior, may inadvertently trigger negative emotions. Child-centric systems should therefore take the time to regularly check on the child’s mental and emotional states and to be ready to provide interventions when necessary. This is where the emotion detection feature of Amy comes in handy. When it detects a negative emotion, it confirms this with the child then proposes two activities - sharing and reflection, and breathing exercise. A guided 5-to-10-minute breathing exercise has been reported by prior studies as effective in improving the emotional well-being of children [18].

Assigning Amy’s roles. Conversational agents have portrayed varying roles based on the context or use cases of the application. These may include serving as a software tutor [8, 11] and a storytelling companion [6], therapist [21] in learning environments, and a caregiver [16] for healthcare systems. Amy is designed primarily as a storytelling companion with a two-fold learning objective: social and emotional. As a social learning companion, Amy utilizes storytelling to impart knowledge about social situations that children may come across with and the skills they would need to respond appropriately. The additional use of pictures to complement the story text, quick replies to guide children towards the correct answer, and non-judgmental corrective feedback were

found to facilitate the learning task, with children retaining lessons particularly for the themes on *taking turns* and *helping others*.

As an emotional learning companion, Amy regularly monitors the child’s emotion, confirms a detected negative emotion, and intervenes with a sharing and reflection session and breathing exercise. However, most children felt more comfortable doing the guided breathing exercise than sharing their experiences with Amy.

5 CONCLUSION

Interactive narrative-based technologies can offer children with ASD the opportunity to learn proper social behavior and emotion regulation in a safe space prior to actual exposure. In this paper, we investigated how interaction with a virtual talking companion, Amy, can prepare these children through narrating of social stories, assessing comprehension of social situations, and regulating emotions through sharing, reflection and breathing exercises. Amy’s conversation flow design and selection of story themes were heavily informed by inputs from parents and caretakers to make children find meaning while staying engaged with the storytelling.

Feedback from parents suggest that future work should look into the generation of emotion-rich speech typically exhibited by human storytellers and to expand the current repertoire of stories. Furthermore, strategies used by therapists and caretakers to elicit story details should be explored to enhance Amy’s current limited ability in encouraging children to share and reflect on their experiences. Longer-term studies with a larger participants base can also shed insights on the usefulness of a virtual storytelling companion in promoting social-emotional learning of children with ASD.

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